

- 1 Fig. 1.1, 1.2 and 1.3 are electron micrographs taken from different cells.

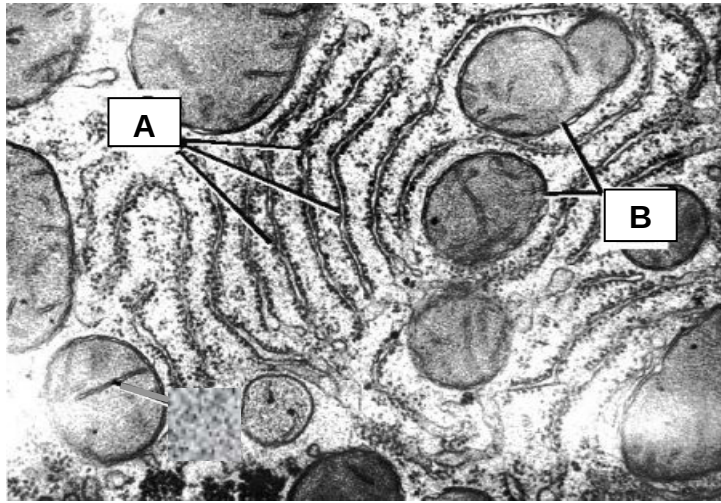


Fig 1.1

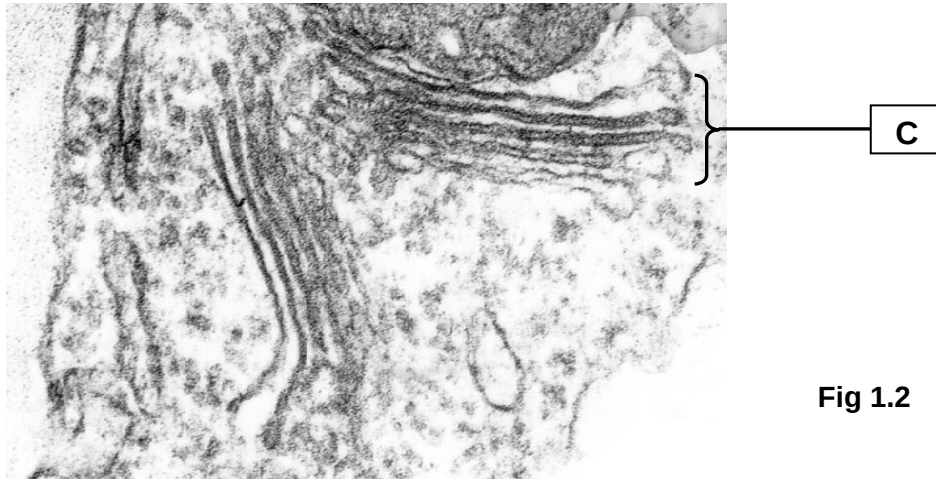


Fig 1.2

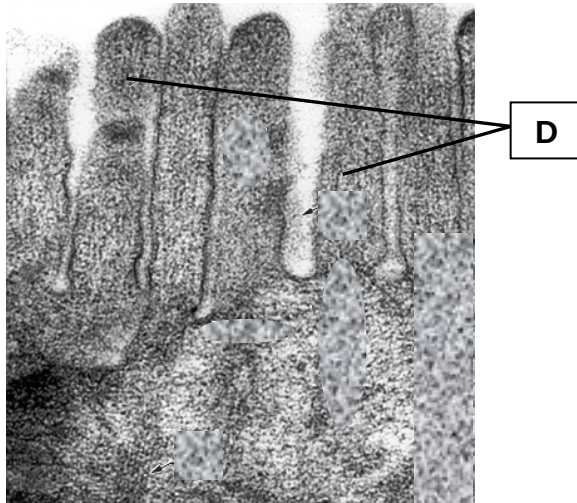


Fig 1.3

(a) Name the structures labeled A to D.

A RER/ ribosomes;

B Mitochondria;

@ ½ mk

C Golgi body/apparatus;

D Microvilli; [2]

[Turn over

(b) Describe briefly how the two structures, **A** and **C**, are related in the cell.

Polypeptides synthesized at ribosomes; are channeled into RER;
where (post-translational) modifications take place / ref. to glycosylation* ;
(*can be mentioned here or with GA)

Intracellular transport via interconnected cisternae;

and (transport) vesicles; to Golgi apparatus;
where final products (eg. ref to lysozyme) are formed;
and packaged / ref. to formation of secretory vesicles/lysosomes; @ ½ mk [3]

ECF – if A labeled as SER, award for synthesis of lipid/ref to SER function; then carry on with the rest of mark scheme

(c) With reference to Fig. 1.1, explain the significance of the numerous **B** found adjacent to **A**.

Synthesis of/provide ATP; R! energy source
required for protein synthesis/translation at ribosome on surface of RER / post-translational modification; @ ½ mk [1]

R! metabolism in general.

(d) (i) State the function of structure **D**.

Increase surface area of cell surface membrane;
for uptake/absorption of material; R! adsorption, exchange; @ ½ mk [1]

(ii) Suggest where such structures may be found in the human body.

small intestine/kidney; @ 1 mk [1]

Fig. 1.4 shows a stalked particle that is found in internal membranes in cells. These structures are responsible for the production of ATP that is driven by the proton (H^+) gradient that occurs across such membranes.

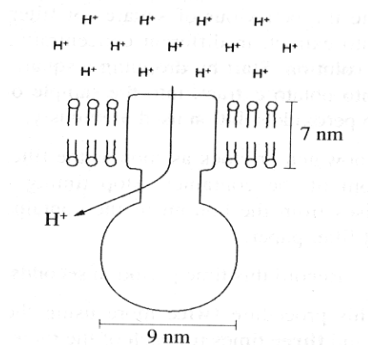


Fig 1.4

(e) State the precise locations of such particles in a general plant cell.

Thylakoid membranes; within chloroplasts;

Cristae/inner membranes; within mitochondria; @ ½ mk [2]

- (f) Describe how the structure of the stalked particle is related to its function.

Stalked particle has hydrophobic portion that allows it to be embedded in the membrane;

Protein (also has a channel; lined with hydrophilic amino acids/ref. to hydrophilic; which allows H^+ to move (along proton gradient); by facilitated diffusion;

Stalked particle contains catalytic/binding sites/ ref. ATP synthase; for binding of ADP;

using proton motive force;

It has 3 parts - rotor, rod, knob;

The rod connects the cylindrical rotor and knob;

it (spins and) activates catalytic activity (on the knob); @ $\frac{1}{2}$ mk [3]

[Total: 13]

- 2 In an experiment to study the effect of heat treatment on the digestibility of protein substrate and the effect of raw bean extract on protease activity, various reaction mixtures were prepared and were incubated for 30 minutes.

The protein concentration of each reaction mixture at the beginning and at the end of the incubation period was determined by the colorimetric method which measures colour intensity of these reaction mixtures. The results were shown in table 2.1.

Table 2.1

Incubation period /min	Colour intensity of the reaction mixture / arbitrary unit		
	Tube A	Tube B	Tube C
	Protease + heated protein substrate	Protease + unheated protein substrate	Protease + unheated protein substrate + raw bean extract
0	10	10	10
30	4	7	9

Fig. 2.1 shows a standard graph obtained by using colorimetric method for determining concentration of protein solutions.

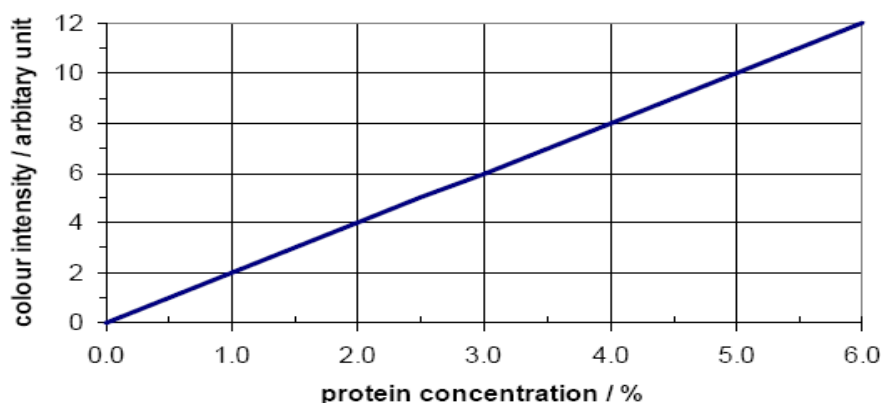


Fig. 2.1

(a) With reference to Table 2.1 and Fig 2.1,

- (i) Determine the protein concentration of the three tubes at time 0 min.

5% protein concentration; @1 mk [1]

- (ii) Explain which protein substrate is more digestible by protease.

heated protein more digestible by protease;

less protein remained after incubation with protease / lowest colour intensity;

unheated = 3.5% protein concentration / decreases by 1.5%;

vs heated protein = 2% / decreases by 3% ; @½ mk [2]

- heating increases the digestibility of this protein substrate / digest more quickly / easier to digest;

@ 1mk

[3]

- @ 1 mk**

[1]

- [illegible]

phospholipid;

@1mk

[1]

- @ 1/2 mk

[2]

- @ 1mk

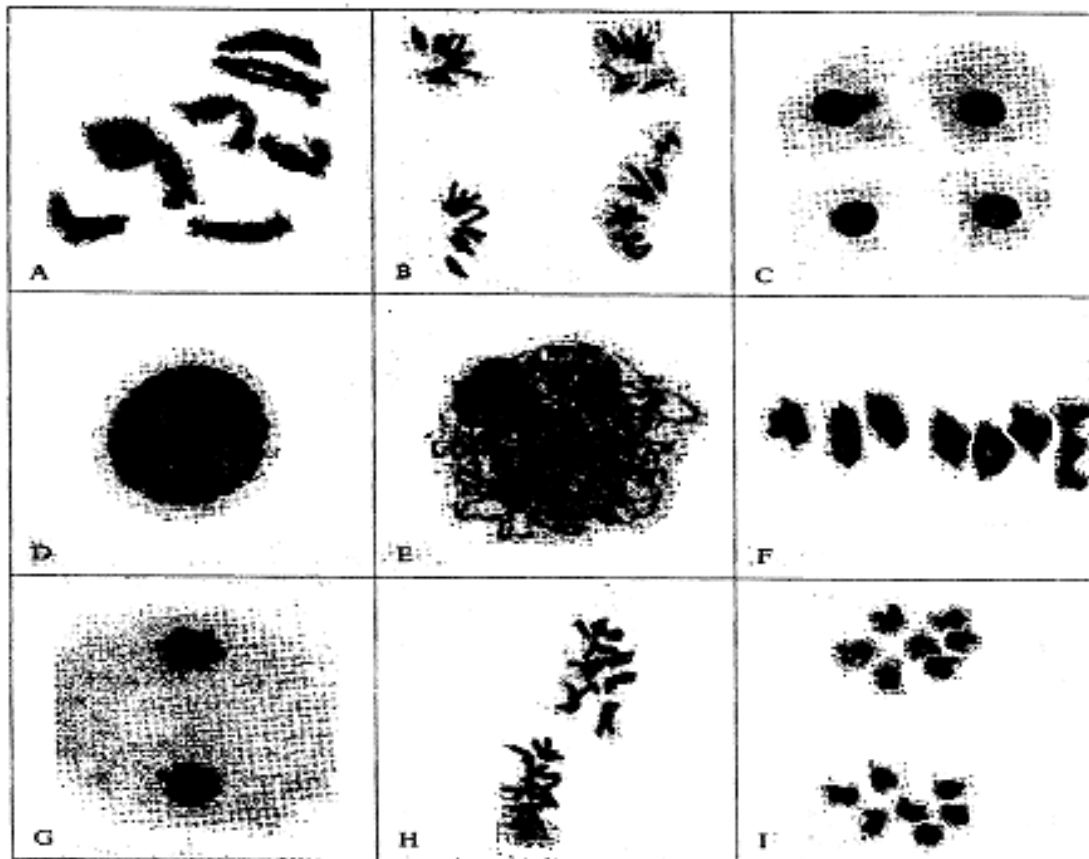
[1]

- oil destroy birds feathers' ability to repel water; birds become waterlogged and drown: @ ½ mk [1

[1]

9747/02/Prelim 08

- 3 The micrographs below show nuclei of cells at various stages during nuclear division in a flowering plant (Fig. 3.1).



Adapted from: *Introducing Genetics*, Jones and Karp, John Murray, 1988.

Fig. 3.1

- (a) Micrograph D shows the first stage in the sequence. In the boxes below, arrange the other letters shown on the micrographs to indicate the correct chronological sequence for this type of nuclear division.

D	E	A	F	I	G	H	B	C
---	---	---	---	---	---	---	---	---

[1]

All must be right for 1mk. No $\frac{1}{2}$ mk.

- (b) State the diploid number of this plant. With reference to **two** stages (name the stages) from the photomicrographs, explain how you arrived at your answer.

Diploid number of this plant = 14;;

Stage A = Prophase I (Both micrograph stage & stage of meiosis must be stated);
7 pairs of homologous chromosomes already paired up from 14 chromosomes;

Stage F = Metaphase I;

7 pairs of homologous chromosomes lined up along equator;

Stage I = Anaphase I / telophase I;

(2 groups of separated homologous chromosomes with each group consisting of)
7 chromosomes at each pole;

@ $\frac{1}{2}$ mk, any 2 stages

Accept explanation if candidate has referred to micrograph or meiosis stage

[3]

- (c) During the stage shown in H, independent assortment will result in an increased variety of gametes if a particular process has taken place at an earlier stage. Name this process and state when this occurs during cell division.

Process: crossing over (at chiasma);
Stage: Prophase I; (A! only if answer is related to synapsis, formation of chiasmata, crossing over) [2]
 @ 1 mk

- (d) With reference to micrograph F in Fig. 3.1, fill in the numbers of the following:

No. of sets of chromosomes	<u>2</u>	No. of centromeres	<u>14</u>
No. of chromatids	<u>28</u>	No. of telomeres	<u>56</u>
No. of DNA strands	<u>56</u>		

- ½ if one is wrong, -1 if two are wrong, - 1½ if three are wrong

- (e) Distinguish between the terms centriole and centromere.

Centriole - an organelle made up of microtubules/proteins;
 ref. to structure eg. cylindrical, occurs in pairs;
 organises spindle fibres / microtubule organising centre / form spindle / ref. to microtubules radiating out from centriole;
 Centriole replicates/divides at interphase;
R! forms spindle fibre/ spindle fibre formation max 2
Centromere: part of a chromosome / made of (noncoding/satellite) DNA;
 holds sister chromatids (of a chromosome together during cell/nuclear division);
 allows kinetochore formation / attachment of spindle fibres; [3]
 Centromere replicates/divides at anaphase; max 2 @ 1mk

- (f) If centromeric DNA is deleted from a chromosome, suggest how this will affect the process of mitosis.

Sister chromatids are not held/joined together;
Kinetochores cannot form on chromosome;
Spindle fibres cannot attach/bind to chromosomes;
chromosomes cannot align along equatorial plane during metaphase;
Ref. to random movement of sister chromatids / chromatids not separated to opposite poles (idea of opposite poles is important) ;
Daughter cells will not have complete set of chromosomes/ have extra or less
chromosomes / will not be genetically identical to parent cell / ref. to aneuploidy; [3]
mitosis may not proceed past metaphase / anaphase cannot occur; @ 1 mk

[Total 14]

- 4 (a) Explain how the promoter of a eukaryotic gene influences transcription.

It is a (regulatory) non-coding (DNA) sequence (found at 5' end of a gene);

allows binding of general transcription factors to TATA box;

allows binding of RNA polymerase (II) ;

to form transcription initiation complex / allows for initiation of transcription;

Promoter works in concert with other (proximal/distal) control elements to control the level of transcription;

@ 1 mk

[3]

- (b) Fig. 4.1 shows the binding of a general transcription factor to part of the promoter. This will help to recruit RNA polymerase to the promoter.

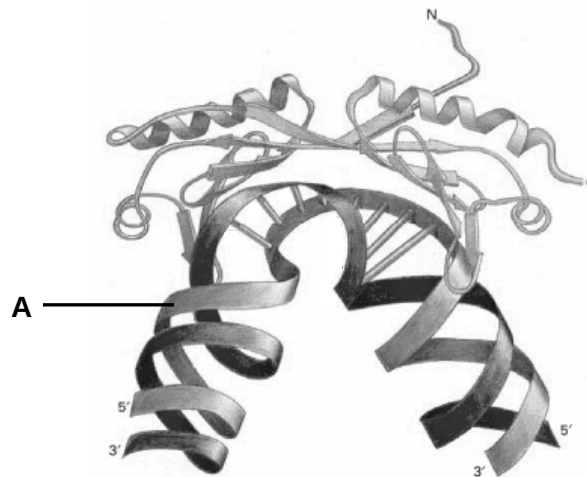


Fig. 4.1

- (i) With reference to Fig. 4.1, give **two** reasons why the structure labelled **A** is part of the promoter and not the transcription factor.

Promoter is made of DNA;

double helix / stranded;

Has 5' and 3' ends;

which are antiparallel; @ 1mk

[2]

- (ii) Suggest why a substitution mutation at the promoter may result in the silencing of the gene controlled by the promoter.

New sequence is not complementary to binding site of / not recognised by transcription factors / RNA polymerase;

Transcription factors / RNA polymerase cannot bind;

Transcription cannot be initiated; @ 1mk

[2]

(d) Describe how distal control elements can help increase the rate of transcription of a gene.

Activators bind to distal control elements / enhancer;

Recruits DNA bending protein/causes DNA to bend;

(More) transcription factors and RNA polymerase brought to promoter;

Form the transcription initiation complex; @ 1 mk [3]

(e) Even though the control of gene expression occurs mostly at the level of transcription, control of gene expression also occurs at the level of post-translational modification.

(i) Name an example of post-translational modification of a **specific** protein.

**eg. insulin; cleavage; / histone; acetylation/methylation; /
protein kinase; phosphorylation;
ubiquitination (no specific eg for this); or any other valid eg. @ ½ mk** [1]

(ii) Suggest the advantage of control of gene expression at the level of translation over control of gene expression at the level of transcription.

Ref. to faster formation of protein (as no time spent in transcription and translation); ½ mk for 'faster' or 'saves time'

possible even if nucleus is absent from cell; @ 1 mk [1]

[Total: 12]

- 5 In *Drosophila* flies, a gene for wing shape found on chromosome 2 has two alleles – allele **A** for curly wing, which is dominant to allele **a** for normal wing.

Female flies are homogametic while males are heterogametic. The gene for *Drosophila* eye colour is located on the X chromosome and has two alleles – the allele for red eyes being dominant to the allele for white eyes.

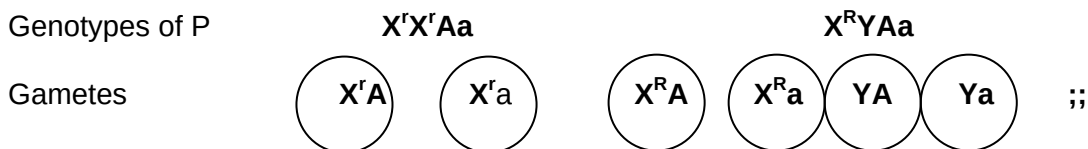
(a) (i) Write the genotypes of the two flies in the table below.

	Fly 1	Fly 2
Sex of fly	Female	Male
Eye colour	White	Red
Wing shape	Curly wing, heterozygous	Curly wing, heterozygous
Genotype	X^rX^rAa ;	X^RYAa ;

[1]

R! using 2 different alphabets for the different alleles of the same gene

- (ii) Draw a genetic diagram showing the phenotypes of the offspring and the expected ratio of phenotypes when these two flies are crossed.



Do not mark on if gametes are wrong

Exception: if gametes shown correctly in Punnett square, no marks for gametes but mark on

F1 genotypes and phenotypes:

	X^RA	X^Ra	YA	Ya
X^rA	X^RX^rAA Red-eyed Curly wing female	X^RX^rAa Red-eyed Curly wing female	X^rYAA White-eyed Curly wing male	X^rYAa White-eyed Curly wing male
X^ra	X^RX^rAa Red-eyed Curly wing female	X^RX^raa Red-eyed Normal wing female	X^rYAa White-eyed Curly wing male	X^rYaa White-eyed Normal wing male

F1 Phenotypic Ratio:

**3 Red-eyed Curly wing female : 1 Red-eyed Normal wing female :
3 White-eyed Curly wing male : 1 White-eyed Normal wing male**

Mark allocation: $\frac{1}{2}$ mk for correct gametes for male,
 $\frac{1}{2}$ mk for correct gametes for female,
 1 mk for correct F1 genotypes
 $\frac{1}{2}$ mk only with one mistake, none for > 1 mistake,
 1 mk for correct F1 phenotypes (*must be associated with correct genotype*)
 $\frac{1}{2}$ mk only with one mistake, none for > 1 mistake,
 1 mk for correct F1 phenotypic ratios,
 none with any mistake or if phenotype is wrong.

[4]

In a separate experiment, when a white-eyed and yellow-bodied (homozygous recessive) female was crossed with a red-eyed and gray-bodied (dominant) male, all the F1 females have gray body and red eyes whereas all the F1 males expressed white eyes and yellow body. In the F2 generation, 98.7% of the total offspring showed the parental phenotypes - yellow-bodied, white-eyed flies and gray-bodied, red-eyed flies. The remaining 1.3% of the flies were either yellow bodies with red eyes or gray bodied with white eyes.

(b) (i) State the possible explanation for the results obtained for F1.

Linkage/the 2 genes found on the X/same chromosome; [1]

(ii) Given the percentages of the different phenotypes obtained in the F2 generation, state the information regarding the two genes investigated that can be calculated.

Crossing over value/ Distance between the 2 genes;

[1]

(iii) State the process and stage during the formation of gametes in the flies that results in the appearance of the above said 1.3% of the offspring.

Crossing over; during Prophase 1 of meiosis;

If answer is 'crossing over during prophase I or ...' award ½ mk

[1]

@ ½ mk

(c) From the results obtained in the above cross, would you conclude that the inheritance of eye and body colour shows continuous or discontinuous variation? Explain your answer.

Discontinuous variation;

Discrete phenotypes with no intermediate/controlled by 1 gene;

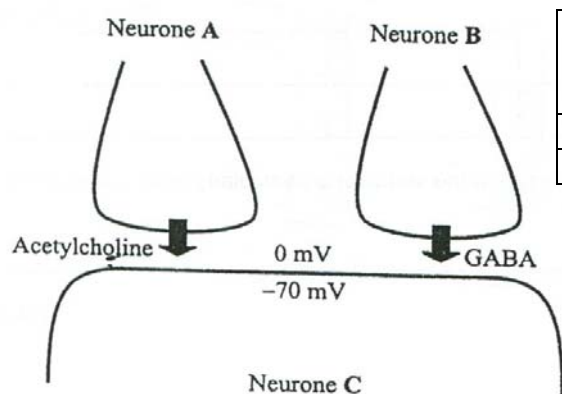
If 'colours are distinct and are controlled by 2 genes', award ½ mk

@1 mk

[2]

[Total: 10]

- 6 Fig. 6.1 shows dendrites from each of two neurones, **A** and **B**, forming synapses with a third neurone, **C**. Concentrations of some ions are also shown.



	Ion concentration / nmol dm ⁻³		
	Na ⁺	K ⁺	Cl ⁻
Outside neurone C	145	12	120
Inside neurone C	12	155	5

Fig 6.1

- (a) (i) Acetylcholine increases the permeability of the postsynaptic membrane to Na⁺ ions. Explain how acetylcholine released from neurone **A** generates an action potential in neurone **C**.

Acetylcholine diffuses across the synaptic cleft;

binds to receptors/chemical-gated Na⁺ channels on neurone C/ postsynaptic membrane;

triggers opening of chemical-gated Na⁺ channels;

causes Na⁺ influx;

If postsynaptic membrane is depolarized till threshold potential, action potential is generated;

@ 1mk, max 4

[4]

- (ii) Neurone **B** releases a substance called GABA as its neurotransmitter. GABA increases the permeability of the postsynaptic membrane to Cl⁻ ions. Suggest how impulses in neurone **B** would reduce the chances of an action potential in neurone **C**.

Chloride ions enter neurone C;

membrane more negative than resting potential / membrane is hyperpolarised;

threshold potential is not reached and action potential not generated in postsynaptic membrane;

more stimulation / impulses / high acetylcholine level from A to generate an action potential;

@ 1mk

[2]

- (iii) Suggest a benefit of neurone **C** synapsing with more than one neurone.

Allow for integration of stimuli from various sources to produce a coordinated response/ ref. to summation / impulse transmission can be better controlled and fine tuned to suit circumstances; OWTTE

@ 1mk [1]

- (b) (i) The acetylcholine receptor is a type of G-protein-linked receptor. When acetylcholine binds to it, a membrane-bound enzyme is activated. Name the enzyme.

Adenylyl cyclase

@ 1mk

[1]

- (ii) Explain the role of the enzyme named in (b)(i) in signal transduction that leads to a response in the postsynaptic neurone.

**(Activated) adenylyl cyclase (A! spelling error) converts ATP;
to cAMP;**

**cAMP acts as a secondary messenger;
activates a protein kinase;**

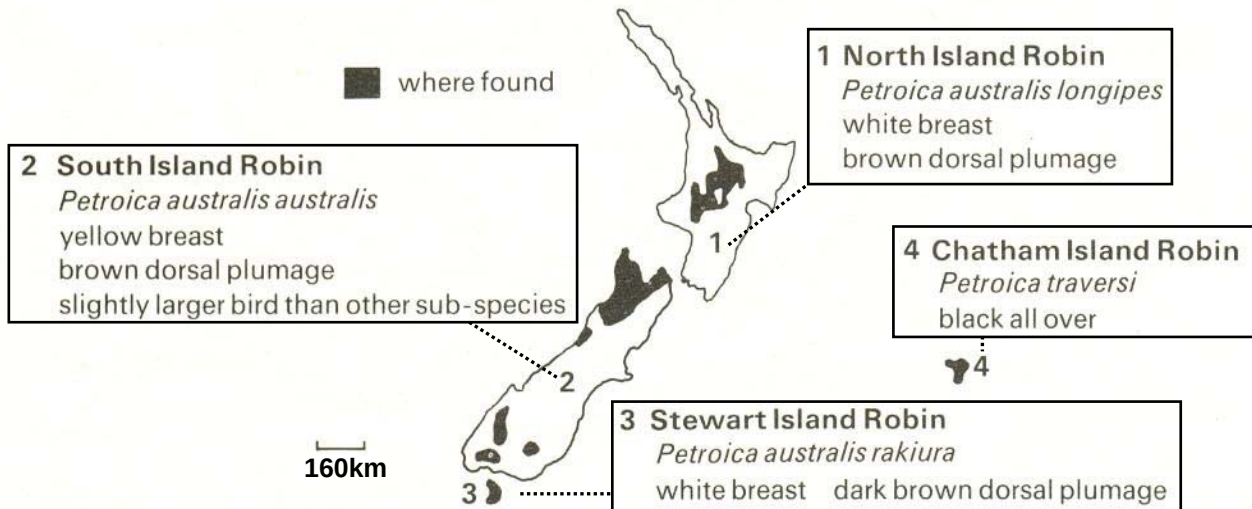
**which activates / phosphorylates other protein kinases/relay proteins/ causes
phosphorylation cascade;**

@ ½ mk, max 2

[2]

[Total: 10]

- 7 In New Zealand today, there are two species and three sub-species of native bush robins. It is believed that the robins evolved from a common ancestral stock, members of which flew from Australia across the Tasman Sea and became established in New Zealand over a million years ago. This ancestral form is considered to be similar to the present day Australian flame robin, *Petroica multicolor*, a bird with a brightly coloured red breast. The New Zealand birds do not have this red colour. Some characteristics of the birds and their distributions are shown below:



- (a) State the scientific name of the two species of native bush robins in New Zealand.

Petroica australis; *Petroica traversi*; (if no underline, ½ mk for both species)@ ½ n [1]

- (b) Explain why the robins in locations 1, 2 and 3 are similar but different from those in location 4.

Birds in locations 1, 2 and 3 are capable of interbreeding;
(because of the) short distances between the islands;

(similar environment and so) subjected to similar selection pressure / gene flow is not interrupted / shared a common gene pool for a long time;

Birds on location 4 could not interbreed with the other birds;

geographically isolated / due to the longer distance between the islands & so do not interbreed/ ;

mutation such as the genes for black plumage would tend to be confined within the isolated inbreeding population;

@ 1mk, max 4

[4]

- (c) Describe how the molecular clock can be used to date evolutionary events, such as when the two species of native bush robins in New Zealand diverged.

Neutral/silent mutations / mutations in non-coding DNA are fixed;
at relatively constant rate;

Ref. to comparing DNA sequences / a particular common gene / amino acid sequence;

Ref. to the number of mutation / substitution / difference in genetic sequence can be used to calculate the length of time since divergence;

@ ½ mk

[2]

- (d) The red breast trait is present in Australian robins. Account for the absence of this trait in New Zealand robins.

Australian robins might have gained the red breast by a mutation (which occurred in the Australian population);

after the colonization of New Zealand by forms which did not possess it / ref to Founder effect;

Red-breasted forms might have reached New Zealand where the characteristic was at a selective disadvantage / vice versa;

Frequency of red breast allele decreased and eventually disappeared from the gene pool;

Ref. to genetic drift such that red breast individuals accidentally die; @ 1 mk [2]

8 (a) Compare and contrast the viral genome and the eukaryotic genome. [8]

	Viral genome	Eukaryotic nuclear genome
Genetic material	1 Either DNA; 2 or RNA;	3 DNA;
	4 Either double-stranded; 5 or single-stranded;	6 Double-stranded;
	7 Either linear; 8 or circular ;	9 Linear;
	10 Either segmented; 11 or single;	12 segmented / many different chromosomes;
Presence of telomeres	13 May or may not be present;	14 Present
Presence of origin of replication	15 May or may not be present;	16 Yes / many present
DNA enclosed in ...	17 capsid;	18 nucleus;
DNA associated with histones	19 No;	20 Yes;
Amount of noncoding DNA	21 Little ;	22 Large amounts;
Amount of repetitive DNA / centromeres	23 Little / Absent	24 Large amounts/ Present
spacer DNA between genes	25 Little	26 Large amounts
Presence of introns in genes	27 No (in most cases);	28 Yes;
Size	29 Smaller;	30 Larger;

@ ½ mk each, award if attempt at valid comparison

8 (b) Explain the role of the F plasmids in bacterial conjugation. [8]

- 1 Conjugation - transfer of genetic material (in one direction) through (direct) cell to cell contact between two bacterial cells;
- 2 Ref to one with the F plasmid (donor) and the other without (recipient);
- 3 F plasmid contains genes / codes for synthesis of sex pilus;
- 4 Used for attachment to recipient cell / formation of mating bridge/conjugation tube;
- 5 Ref. to F factor replicates by rolling-circle mechanism (*idea of replication is impt*);
- 6 (single-strand of) F plasmid is transmitted through mating bridge to recipient;
- 7 transferred ssDNA is replicated in the recipient cell, producing a ds F plasmid;
- 8 Conjugation tube breaks;
- 9 (if entire F plasmid was transferred) Recipient cell now has F plasmid/ is F⁺;
- 10 F plasmid can integrate into bacterial chromosome, forming Hfr cell;
- 11 Some genes from host chromosome are also transferred to recipient;
- 12 conjugation tube usually breaks before the entire F plasmid is transferred; @ 1 mk

8 (c) Suggest the advantages of conjugation as compared to transduction. [4]

- 1 No need for 3rd party / virus / vice versa;
- 2 Host cell remains alive;
- 3 Does not rely on chance that wrong DNA fragments are packaged (in generalised transduction);
- 4 Does not rely on chance that viral DNA excises imprecisely (in specialised transduction);
- 5 Homologous recombination not needed unlike (general) transduction (though specialised transduction does not require it);
- 6 Higher chance to transfer genetic material (if cells contact each other) (as transduction relies heavily on chance);
- 7 In transduction, genetic material may or may not be degraded / vice versa;
- 8 Larger chunk of DNA/genetic material can be transferred;
- 9 Ref. defined genes (of F plasmid) transmitted during conjugation vs random DNA fragments (via general transduction) OR only genes/DNA fragments beside prophage (via specialised transduction) @ 1 mk

9 (a) Explain the principles of homeostasis in terms of receptors, effectors and negative feedback. [7]

- 1 definition: maintenance of a constant internal environment;
 - 2 organism not adversely affected by external environment;
 - 3 optimal level known as reference point/set point;
 - 4 receptor/detector detects signal / change/deviation (from reference point);
 - 5 information relayed to control centre;
 - 6 (control centre is) where information compared with reference point
 - 7 control centre initiates response;
 - 8 effector returns system to optimal;
 - 9 by eliminating deviation/producing opposite effect;
 - 10 (linked to point 9) ref. to negative feedback;
- @ 1mk

9 (b) Compare and contrast competitive and non-competitive inhibitors and their effects on the rate of enzyme activity. [7]

- 1 Both serve to lower rate of enzyme activity
- 2 Both are reversible @1 mk

Competitive	Non-competitive
3 Structural similarity to substrate	4 No structural similarity to substrate
5 Binds to active site	6 Binds to site other than active site
7 Competes with substrate for active site	8 Does not compete with substrate for active site
9 Prevents/lowers formation of ES complex	10 May or may not prevent formation of ES complex
11 Does not change conformation of enzyme upon binding	12 Changes conformation of enzyme upon binding
13 Increasing [S] concentration alleviates effect of inhibitor	14 Increasing [S] concentration does not alleviate effect of inhibitor
15 Max rate of reaction can be reached (with increased S concentration) / $V_{\max}$ unchanged	16 Max rate of reaction cannot be reached (with increased S concentration) / $V_{\max}$ reduced

@ ½ mk, award if attempt at valid comparison

9 (c) Suggest similarities between the interaction of substrate with an enzyme and the interaction of ligand with a receptor. [6]

- 1 Both bind to specific regions of protein (idea of specific region/portion is imp);
 - 2 Active site for enzyme;
 - 3 Binding site for receptor;
 - 4 Both are complementary in shape to sites that they bind to;
 - 5 Both will bind to protein via H bonds, ionic bonds, hydrophobic interactions;
 - 6 Both can induce a conformational change in the protein when they bind;
 - 7 Ref. to induced fit hypothesis for enzymes;
 - 8 Ref. to activation of receptor;
 - 9 Both interactions are not permanent;
 - 10 Ligand dissociates from binding site of receptor / Ligand-receptor complex are removed;
 - 11 ES complex formed will be converted to product, which is released;
- @ 1 mk